

Reproducing the Speed, Utility, and Fairness of TABFAIRGDT

COMP5331 project type: Implementation

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Declaration: This project is conducted solely within COMP5331. It is not work for another course, a research project, or an FYP.

Project description

Motivation and objective

Synthetic tabular data can support data sharing, augmentation, and controlled evaluation, but useful synthetic data should preserve more than marginal feature distributions. If a generator reproduces biased relationships between a sensitive attribute and a prediction target, models trained on its samples may inherit unequal treatment. Conversely, a generator that achieves a favorable fairness measure by erasing information may damage downstream utility. TABFAIRGDT, proposed by Panagiotou and colleagues, uses autoregressive decision trees to generate fair tabular data quickly. Our implementation project will evaluate its three linked claims—speed, predictive utility, and fairness—under one coherent protocol on a small selection of public datasets.

The key question is how TABFAIRGDT's utility, fairness, and runtime change with the strength of its fairness intervention when the same downstream learner and held-out real test data are used throughout. We will compare the released generator at $\lambda=0$ (no fairness adjustment), $\lambda=0.5$, and $\lambda=1$, alongside a classifier trained on the real training set. This design isolates the intervention within one generator while retaining the paper's three principal evaluation dimensions.

Relation to fair-data literature

FairGAN establishes fair synthetic generation as a way to influence the behavior of classifiers trained on generated records, and provides a useful conceptual comparison to a tree-based generator. Panagiotou, Roy, and Ntoutsis's comparative study examines the joint effects of target-class and sensitive-group imbalance, directly motivating our

group-count diagnostics. Abroshan and colleagues distinguish fairness constraints imposed on generated data from the behavior of downstream predictions; this helps us specify which notion the TABFAIRGDT experiment actually tests. We will use these readings to define the fairness metrics and interpret the utility trade-off. The numerical comparison centers on the released tree generator's unadjusted and fairness-adjusted settings; FairGAN informs the interpretation of related generative approaches.

Paper, code, and datasets

The principal paper is Emmanouil Panagiotou and coauthors, "TABFAIRGDT: A Fast Fair Tabular Data Generator using Autoregressive Decision Trees," ICDM 2025. The author repository includes a usage example, main experiment script, dependency list, and evaluation or baseline modules. We will pin a commit and first run the provided small example to verify that generation and evaluation operate in our environment. We will then trace which repository settings correspond to the fairness intervention described in the paper. If the published implementation contains multiple parameterizations, we will choose one documented setting for the main comparison and reserve a small parameter sweep for analysis.

We will use the Adult configuration in the author repository as the primary tabular classification dataset. Its loader points to an externally hosted cleaned file named `adult-clean.csv`, not directly to the raw UCI Adult download. We will record that file's source and version, compare its fields and class counts with the UCI description, and specify the target label, sensitive attribute, missing-value treatment, category encoding, and train/test split. We will fix the real-data test set before fitting any generator. All imputation, scaling, and category vocabularies used to train the generator or downstream model will be learned from the real training partition only. This separation prevents information from the held-out test set entering synthetic training data and makes downstream comparisons meaningful.

Reproduction checkpoints

The released repository offers a minimal example and separate experiment scripts; its dataset loader can retrieve supported benchmarks and cache them locally. We will first use the smallest documented example to identify the generator's fitting and sampling calls. Adult will be the first full benchmark because it has a clearly documented prediction label and a commonly used group attribute. We will record the loader's raw source, preprocessing choices, cache version, and train/test split, then repeat the same downstream classifier evaluation for real and generated training samples. A utility result will be accompanied by the synthetic data's group counts and outcome prevalence, since a fairness score can change simply because one group is undersampled. The baseline set will be the no-generation real-data reference and TABFAIRGDT with $\lambda=0$; $\lambda=0.5$ and $\lambda=1$ provide the fairness-adjusted conditions. All generated conditions will have matched sample counts and preprocessing.

Evaluation protocol

Our main comparison will include a downstream classifier trained on the real training set and the same classifier trained on equally sized synthetic samples generated at $\lambda=0$, 0.5 , and 1 . The released example documents $\lambda=0$ as the unadjusted CART setting. We will use the same downstream classifier family and tuning budget for every generated training set. We will repeat experiments under several seeds, using saved real splits so that the fairness intervention is the principal changing factor.

On the held-out real test set, we will measure predictive accuracy and, where probabilities are meaningful, AUROC or macro-F1. For fairness, we will report demographic parity difference and equal opportunity difference by the chosen sensitive attribute, with the group denominators shown. These metrics answer different questions: parity concerns the rate of positive predictions, while equal opportunity conditions on true positives. Neither alone establishes an unqualified notion of fairness. We will present both, together with utility, so that one cannot mask a severe trade-off. We will also measure generator fitting time, sampling time per fixed number of records, and downstream training time. To investigate whether fairness improvements arise from dataset distortion, we will

compare selected numeric distributions, category frequencies, target prevalence, and the number of synthetic records in each sensitive group.

The principal analysis will use the released `lamda` parameter at 0, 0.5, and 1. We will plot utility against each fairness metric and annotate runtime, forming a compact trade-off picture rather than reporting a single favorable configuration. We will compare variability across seeds and note whether a group has too few test observations for a stable equal-opportunity estimate. In that case, we will report the count and uncertainty rather than treating a small numerical difference as reliable. We will also inspect whether a generator changes the base rate of the target in a way that makes downstream performance look fair only because the prediction problem became easier or less informative.

Deliverables and expected insight

To verify the evaluation chain, we will examine a small synthetic sample before computing headline metrics. We will check column types, allowed category values, missing entries, duplicate frequency, and whether the sensitive attribute and label appear in the expected format. We will then confirm that the downstream test rows never enter generator fitting and that the same test observations are scored for every method. Finally, the benchmark table will include sample counts and hardware details alongside wall-clock times, since isolated speed numbers are difficult to interpret without workload context. These checks make the fairness comparison reproducible and help locate failures before they become apparently meaningful results.

The project will provide scripts for data preparation, generator fitting and sampling, downstream evaluation, and plots, together with the random seeds and generated samples used for every reported setting. The final report will relate the observed results to the paper's claims and separate numerical replication from any reduced-scope experiment. We expect to identify settings in which the tree-based approach is computationally attractive, and to learn whether fairness gains survive evaluation on untouched real data without an unacceptable loss of predictive utility. Mixed results are also scientifically useful: they can show that a particular dataset, classifier, or fairness definition changes the apparent ranking. The proposed workload is feasible on CPU hardware while retaining enough of the paper's evaluation chain to make the reproduction substantive.

Papers to read

1. **Main paper:** Emmanouil Panagiotou, Benoît Ronval, Arjun Roy, Ludwig Bothmann, Bernd Bischl, Siegfried Nijssen, and Eirini Ntoutsi. "TABFAIRGDT: A Fast Fair Tabular Data Generator using Autoregressive Decision Trees." *IEEE ICDM*, 2025. [arXiv paper](#).
2. **Generative baseline and motivation:** Depeng Xu, Shuhan Yuan, Lu Zhang, and Xintao Wu. "FairGAN: Fairness-aware Generative Adversarial Networks." *IEEE BigData*, pp. 570–575, 2018. [arXiv paper](#).
3. **Imbalance evidence:** Emmanouil Panagiotou, Arjun Roy, and Eirini Ntoutsi. "Synthetic Tabular Data Generation for Class Imbalance and Fairness: A Comparative Study." arXiv:2409.05215, 2024. [Preprint](#).
4. **Fairness definition:** Mahed Abroshan, Andrew Elliott, and Mohammad Mahdi Khalili. "Imposing Fairness Constraints in Synthetic Data Generation." *AISTATS*, PMLR 238:2269–2277, 2024. [Proceedings paper](#).